8 Variations and extrema of a function

Sheet 43 Thu 11 Feb. 2027 *Increasing, decreasing, monotonicity*

1. Warm-up · Quick [IMG] $f(x) = -2x + 5$. Calculate $f(1)$ and $f(3)$; what do you notice?

Ans. $f(1) = 3$, $f(3) = -1$: the order of the images is the reverse of that of the inputs (f is decreasing).

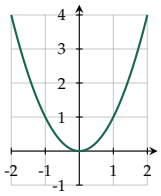
2. Core · MCQ [MOD] f is increasing on I means that, for all $a < b$ in I:

A. $a \leqslant b$ only **B.** $f(a) \geqslant f(b)$ **C.** $f(a) \leqslant f(b)$ **D.** $f(a) = f(b)$

Ans. C. $f(a) \leqslant f(b)$

Increasing: preserves the order; decreasing: reverses it.

3. Core + · Graph [REG] Curve of $f(x) = x^2$ on $[-2, 2]$:



Give the variations of f.

Ans. Decreasing on $[-2, 0]$, increasing on $[0, 2]$.

Expression used to draw the curve: $f(x) = x^2$.

4. Review · Quick [PRO] Which multiplier corresponds to an increase of 15%? To a decrease of 15%?

Ans. 1.15; 0.85.

5. Thread · Quick [DIV] Is 457 divisible by 3? By 9? Justify.

Ans. Sum of digits $= 16$, not a multiple of 3: divisible by neither 3 nor 9.

Sheet 44 Mon 1st Mar. 2027 *Table of variations; maximum and minimum on an interval*

1. Warm-up · Quick [IMG] Calculate x^2 for $x = -2$, 0, 3.

Ans. 4; 0; 9.

2. Core · Method [REG] f is decreasing on $[-3, 1]$ with $f(-3) = 5$, $f(1) = -1$, then increasing on $[1, 4]$ with $f(4) = 4$. Give the maximum and the minimum of f on $[-3, 4]$.

Ans. Maximum 5 reached at $x = -3$ ($5 > 4$); minimum -1 reached at $x = 1$.

Compare the values at the endpoints of the interval AND at the changes of variation.

3. Core + · True/false [REG] True or false? Justify. The maximum of a function on $[a, b]$ is always reached at a or at b.

Ans. False. Counterexample: $f(x) = -x^2 + 4$ on $[-2, 2]$: maximum 4 reached at 0.

4. Review · Table [PNP] Is this table a proportionality table?

x	1	2	3
y	5	9	13

Ans. No: $\frac{5}{1} \neq \frac{9}{2}$ (constant increase $+4$ but $y(0) \neq 0$).

5. Thread · Method [GEO] Is the triangle with sides 5, 12 and 13 right-angled? Which property should be used?

Ans. $5^2 + 12^2 = 169 = 13^2$: converse of Pythagoras' theorem, right-angled (right angle opposite the side 13).

Sheet 45 Tue 2 Mar. 2027 *Affine functions: rate of change and variations*

1. Warm-up · Quick [IMG] $f(x) = 3x - 1$. Calculate $\dfrac{f(4) - f(1)}{4 - 1}$.

   Ans. $\dfrac{11 - 2}{3} = 3$: this is the slope m.

2. Core · Quick [MOD] Direction of variation of $f(x) = -4x + 7$?

   Ans. $m = -4 < 0$: f is strictly decreasing on $\mathbb{R}$.

3. Core + · Proof [LIT] Show that $f(x) = 5x - 2$ is increasing on $\mathbb{R}$.

   Ans. For $a < b$: $f(b) - f(a) = 5(b - a) > 0$, so $f(a) < f(b)$.

   Model write-up for the proof required by the syllabus.

4. Review · Table [LEC] Ages of the members of a club.

Age	10	11	12	13
Frequency	3	6	10	6

   How many members are at most 11 years old? What relative frequency is that?

   Ans. $3 + 6 = 9$ members; $\dfrac{9}{25} = 36\%$.

5. Thread · Programme → formula [REG] Programme: multiply by 3 then subtract 2. Give $f(x)$ then $f(4)$.

   Ans. $f(x) = 3x - 2$; $f(4) = 10$.

Sheet 46 Thu 4 Mar. 2027 *Absolute value and square functions: sign and variations*

1. Warm-up · Quick [CAL] Calculate $|-5|$, $(-5)^2$ and $|2 - 7|$.

   Ans. 5; 25; 5.

2. Core · Method [REG] Give the table of variations of $x \mapsto |x|$ and its sign.

   Ans. Decreasing on $(-\infty, 0]$, increasing on $[0, +\infty)$; minimum 0 at 0; always non-negative.

3. Core + · Reasoning [CMP] Compare $(-3)^2$ and $(-2)^2$ then 3^2 and 2^2. Is the square function increasing on $\mathbb{R}$?

   Ans. $9 > 4$ in both cases, but $-3 < -2$: squaring reverses the order on $(-\infty, 0]$; no, it is not increasing on $\mathbb{R}$.

4. Review · Method [PRB] Two fair dice are rolled. Probability that the sum is 7?

   Ans. 6 favourable cases out of 36: $\dfrac{1}{6}$.

5. Thread · Method [ALG] Programme: choose a number; add 1; square it; subtract the square of the starting number. What is the result for 6? What is the general expression?

   Ans. For 6: $49 - 36 = 13$. In general $(x + 1)^2 - x^2 = 2x + 1$.

Sheet 47 Mon 8 Mar. 2027 *Reciprocal function: variations; relative position of $y = x$ and $y = x^2$*

1. Warm-up · Quick [CAL] Calculate $\dfrac{1}{2}$, $\dfrac{1}{-4}$ and $\dfrac{1}{0.1}$.

   Ans. 0.5; -0.25; 10.

2. Core · Method [CMP] Compare $\dfrac{1}{3}$ and $\dfrac{1}{5}$; what can be said about the variations of the reciprocal function on $(0, +\infty)$?

   Ans. $\dfrac{1}{3} > \dfrac{1}{5}$ although $3 < 5$: the reciprocal function is decreasing on $(0, +\infty)$ (and on $(-\infty, 0)$, but not on $\mathbb{R}^*$).

3. Core + · True/false [CMP] True or false? Justify. For every real number $x > 0$, $x^2 \geqslant x$.

   Ans. False. Counterexample: $x = \dfrac{1}{2}$, $x^2 = \dfrac{1}{4} < \dfrac{1}{2}$. In fact $x^2 \geqslant x$ if and only if $x \geqslant 1$ (for $x > 0$).

4. Review · Quick [PER] What are the front view and the top view of a cylinder standing on its base?
   Ans. Front: a rectangle; top: a disc.
5. Thread · Prove [LIT] Show that the product of two even integers is a multiple of 4.
   Ans. With $a = 2p$, $b = 2q$: $ab = 4pq$, a multiple of 4.

Sheet 48 Tue 9 Mar. 2027 *Comparing $f(a)$ and $f(b)$; optimisation problems*

1. Warm-up · Quick [CMP] $f(x) = x^2$. Compare $f(-3)$ and $f(2)$.
   Ans. $f(-3) = 9 > 4 = f(2)$.
2. Core · Method [REG] $f(x) = x^2$, $a = -1.5$ and $b = -2$. Compare $f(a)$ and $f(b)$ without calculating, using the variations.
   Ans. $-2 < -1.5 \leqslant 0$ and f is decreasing on $(-\infty, 0]$: $f(-2) > f(-1.5)$. Check: $4 > 2.25$.
3. Core + · Optimisation [MOD] $A(x) = x(10 - x)$ for $x \in [0, 10]$ (area of a rectangle with perimeter 20). Show that $A(x) = 25 - (x - 5)^2$ and deduce the maximum.
   Ans. $25 - (x-5)^2 = 25 - x^2 + 10x - 25 = x(10-x)$; maximum 25 reached for $x = 5$ (square of side 5).
4. Review · Quick [STA] 18 pupils out of 24 have an average above 10. Give the relative frequency as a percentage.
   Ans. $\dfrac{18}{24} = 75\%$.
5. Thread · MCQ [EQ] The solution of $\dfrac{x}{3} + 1 = \dfrac{7}{2}$ is:

   A. $\dfrac{15}{2}$ **B.** $\dfrac{9}{2}$ **C.** $\dfrac{21}{2}$ **D.** $\dfrac{5}{6}$

   Ans. A. $\dfrac{15}{2}$

   Pitfall: forgetting to subtract the 1 before multiplying by 3 ($\frac{21}{2}$).